Forecasting Future Realized Volatility

Classical, Tree-Based, and Sequence Models

Chong Gu
Indiana University Bloomington

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Problem Setup

Goal

Predict next 5-day annualized realized volatility for a panel of U.S. stocks and ETFs.

$$FVOL_{i,t}^{(5)} = \sqrt{\frac{252}{5}\sum_{j=1}^{5} r_{i,t+j}^2}$$

- More persistent and economically meaningful than price direction.
- Directly useful for risk management and derivative-related analysis.

Data and Features

Data

- 55 equities/ETFs
- daily OHLCV, 2016–2025
- train: 2016–2024, test: 2025

Proxies

- Market: SPY, QQQ, IWM, VTI
- Macro/credit: VIX, Treasury yields, HYG, LQD

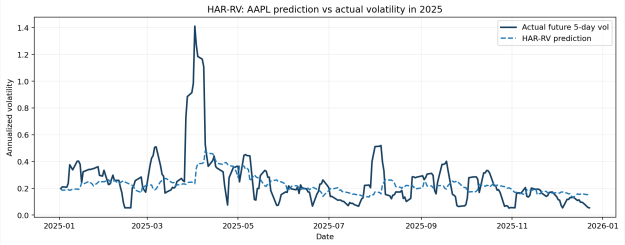
Feature groups

- realized and historical volatility
- high-low range and Parkinson volatility
- volume and liquidity surprise
- market and macro regime signals

HAR-RV Benchmark

$$\log FVOL_{i,t}^{(5)} = \alpha_i + \beta_1 X_{i,t}^{(1d)} + \beta_2 X_{i,t}^{(5d)} + \beta_3 X_{i,t}^{(21d)}$$

- clean financial baseline
- interpretable coefficients
- weak point: lag during regime changes

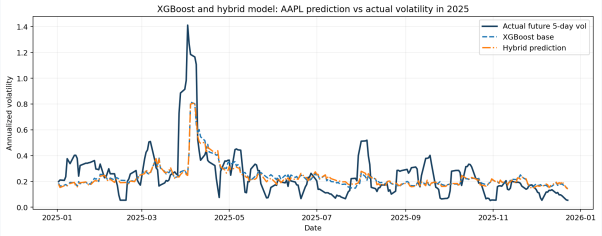


Test on 2025 data: RMSE = 0.1709, MAE = 0.1010

XGBoost and Hybrid Residual Model

Model	RMSE	MAE
XGBoost	0.1598	0.0951
Hybrid	**0.1584**	**0.0939**

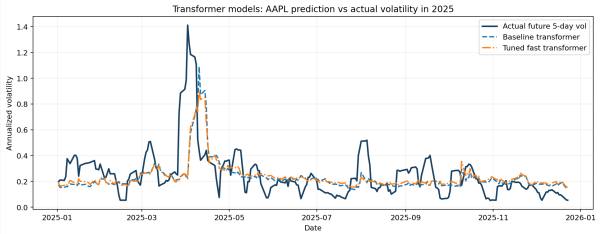
- XGBoost is the strongest standalone tabular model.
- Residual TCN learns what XGBoost still misses.
- Hybrid is the best overall model in the final comparison.



Transformer Baseline and Tuned Variant

Model	RMSE	MAE
Baseline	0.1679	0.0989
Tuned fast	**0.1642**	**0.0966**

- Sequence modeling can improve after tuning.
- Fast features and stronger regularization helped.
- Still weaker than the XGBoost-based line.



Overall Comparison

Model	RMSE	MAE	QLIKE
HAR-RV	0.1709	0.1010	-1.1762
Baseline transformer	0.1679	0.0989	-1.2930
Tuned transformer	0.1642	0.0966	-1.2613
XGBoost	0.1598	0.0951	-1.3433
XGBoost + residual TCN	**0.1584**	**0.0939**	**-1.3569**

- Best classical benchmark: **HAR-RV**
- Best deep model: **tuned transformer**
- Best overall model: **XGBoost + residual TCN**

Remarks

- HAR-RV gives a rigorous financial benchmark.
- A residual sequence model can slightly improve the best tree model.
- Transformers are informative, but still more sensitive to lag and low-vol bias.

Thank you!